

Intermediate Elective

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Set Up

Packages

```
# general use
library(tidyverse)
library(snakecase)
library(janitor)

# file organization
library(here)

# reading .xlsx files
library(readxl)

# data visualization
library(naniar)
library(patchwork)
library(gghighlight)
library(paletteer)
library(ggplot2)
```

Data

```
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")
```

Problems

1. Explain your inspiration

I was inspired by Tobias Stalder's "Hiking Locations in Washington" because the figure's characteristics would allow me to show average abundance/L data for a specific aquatic invertebrate species, separated by NCOS location, while also highlighting a mean average abundance/L value for each site.

This figure makes sense for the aquatic invertebrate data set because it allows for a clear comparison of average abundance/L between different sites, just as the original figure is separated by hiking locations.

The plot will show average abundance/L of ostracods across NCOS locations, the x axis will be NCOS location and the y axis will be average abundance/L. Each jitter point will represent a different average abundance/L measurement, and the mean average abundance/L for each location will be highlighted with a different color to stand out.

2. Sketching Figure

[ViewPDF](#)

3. Code your figure

Cleaning

```

taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))

# creating new object
ncos_sites <- sites |>
# filtering to only include ncos quarterly sampling sites
filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,

```

```

    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
# liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
         date_site,
         med_ph,
         med_sal,
         med_do),
         by = "date_site") |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek")) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
= TRUE),
       site_full = fct_rev(site_full))

```

Wrangling

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
  # filter to only ostracods
  filter(taxon == "ostracod") |>
  # remove missing abundance/L values
  filter(!is.na(ave_lit)) |>
  # group by site
  group_by(site_full) |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # ungroup data frame
  ungroup()
```

Plotting

```
# setting limits for ring plot, max values
y_max <- max(ost_abundance$log_ave_lit, na.rm = TRUE)
ring_log <- pretty(c(0, y_max), n = 4)

# creating new plot
ostracods_plot <- ggplot(
  # inserting ostracod data frame
  data = ost_abundance,
  # setting x axis as site and y axis as average abundance/L
  mapping = aes(
    x = site_full,
    y = log_ave_lit)) +
  # setting ring geometry, color
  geom_hline(
    yintercept = ring_log,
    color = "lightgray") +
  # setting geometry for line for each site
  geom_vline(
    xintercept = 1:length(unique(ost_abundance$site_full)),
    linetype = "dashed",
    color = "gray30") +
  # setting geometry for jitter data points, size and shape
  geom_jitter(
```

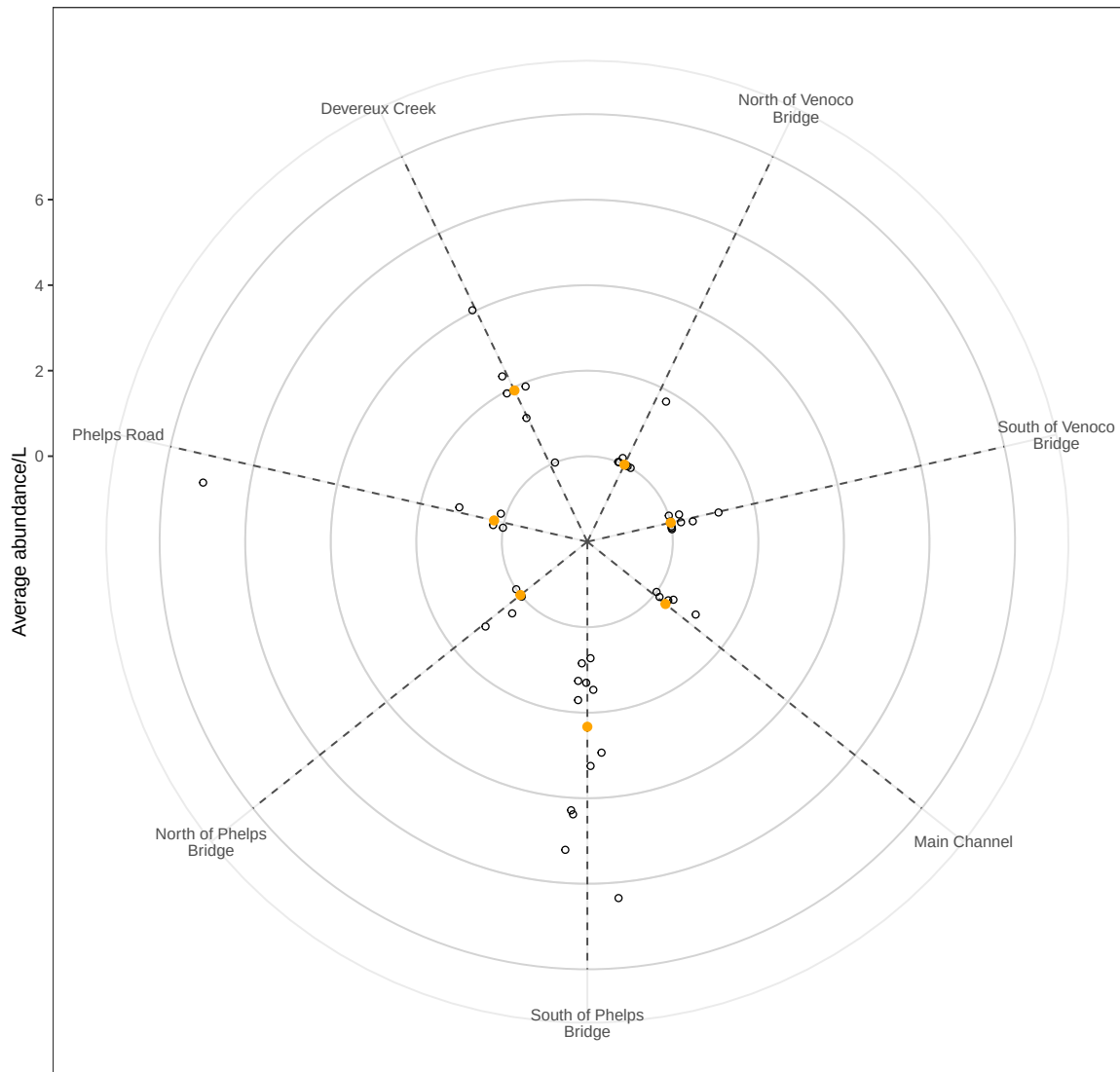
```

    height = 0,
    width = 0.1,
    shape = 21) +
# creating mean average abundance/L data point in different color
stat_summary(
  geom = "point",
  fun = "median",
  size = 2,
  color = "orange") +
# displaying site names on x axis
scale_x_discrete(labels = \(x) str_wrap(x, 15)) +
#scale y axis so bars don't start in the center
scale_y_continuous(limits = c(-2, 8),
  expand = c(0,0),
  breaks = c(0, 2, 4, 6))+
# labeling axes and setting title
labs(x = "",
  y = "Average abundance/L",
  title = "Ostracod Abundance at NCOS During 2025 Water Year") +
# turning plot into a circle (polar coordinates)
coord_polar() +
# changing theme
theme_bw()

#displaying plot
ostracods_plot

```

Ostracod Abundance at NCOS During 2025 Water Year



4. Describe your process

Using someone else's code to create a visualization was helpful in many ways to create the general shape of the plot, but in other ways it also made it more difficult for me to make my own edits and customizations to fit this unique data set because I was less familiar with how the geometries and scales were set up, having not coded them myself.

The main challenge I encountered as I made my visualization was everything to do with the

axes. I couldn't get the plot to keep the scale of the y axis I wanted, and I couldn't get the axis markers onto the actual circle lines like I intended or like in the inspiration plot by Stalder.

This visualization is a spin on a typical jitter plot but the x axis, in this case site, is curved into a circle which flares the data points outwards in different directions for each site. It is most similar to the plot I made in Individual Assignment 5, with major transformations to the axes.